

$$[\neg P \vee (\neg Q \wedge P)] \wedge [\neg Q \wedge (\neg P \rightarrow Q)]$$

$$[\neg P \vee (\neg Q \wedge P)] \wedge [\neg Q \wedge (P \vee Q)], \text{ID}$$

$$[(\neg P \vee \neg Q) \wedge (\neg P \vee P)] \wedge [(\neg Q \wedge P) \vee (\neg Q \wedge Q)] \quad \text{distributive}$$

$$[(\neg P \vee \neg Q) \wedge \neg P] \wedge [(\neg Q \wedge P) \vee F_0]$$

$$(\neg P \vee \neg Q) \wedge F_0$$

$$\neg P \vee \neg Q$$

$$\neg (P \wedge Q)$$

$$P \wedge F_0 \Rightarrow F_0$$

$$P \vee F_0 = P$$

$$(P \rightarrow \neg Q) \wedge \{ [\neg (R \vee \neg P) \wedge (Q \vee P)] \vee (R \wedge P) \}$$

$$(\neg P \vee \neg Q) \wedge \{ (\neg R \wedge \underline{P}) \wedge (Q \vee P) \vee (R \wedge P) \}$$

$$(\neg P \vee \neg Q) \wedge \{ [(\neg R \wedge \underline{P}) \vee (R \wedge \underline{P}) \vee (Q \vee P)] \}$$

$$(\neg P \vee \neg Q) \wedge \{ P \wedge (\neg R \wedge R) \vee (Q \vee P) \}$$

$$(\neg P \vee \neg Q) \wedge \{ (P \wedge F_0) \vee (Q \vee P) \}$$

$$(\neg P \vee \neg Q) \wedge \{(Q \wedge F_0) \vee (Q \vee P)\}$$

$$(\neg P \vee \neg Q) \wedge \{F_0 \vee (Q \vee P)\}$$

$$(\neg P \vee \neg Q) \wedge (Q \vee P)$$

$$\neg (R \vee \neg P) \wedge (Q \vee P) \vee (R \wedge P)$$

$$(\neg R \wedge \neg P) \wedge (Q \vee P) \vee (R \wedge P)$$

$$\downarrow$$

$$[(\neg R \wedge \neg P) \vee (R \wedge \neg P) \wedge (Q \vee P)]$$

$$P \wedge (\neg R \vee R) \wedge (Q \vee P)$$

$$(P \wedge \neg R) \vee (P \wedge R)$$

$$(P \wedge \neg Q) \wedge (\neg R) \vee (P \wedge \neg R)$$

$$P \vee (\neg Q \wedge R \vee \neg R)$$

$$A = \{1, 2, 3\} \quad B = \{2, 3, 4\}$$

$$A \cup B = \{1, 2, 3, 4\}$$

$$A \cap B = \{2, 3\}$$

$$A - B = \{1\}$$

$$A \Delta B = A \cup B - A \cap B$$

$$A \Delta B = \{1, 4\}$$

$$A = \{1, 3, 5\} \quad U = \{\cancel{1}, \cancel{2}, 3, 4, 5, 6, 7\}$$

$$B = \{2, 3, 5, 6\}$$

$$C = \{5, 6, 7\}$$

$$\overline{A \Delta B} \cap C \quad A \Delta B = A \cup B - A \cap B$$

$$A \cup B = \{1, 2, 3, 5, 6\}$$

$$A \cap B = \{3, 5\}$$

$$A \Delta B = \{1, 2, 6\}$$

$$\overline{A \Delta B} = A \Delta B - U$$

$$C = \{5, 6, 7\}$$

$$\overline{A \Delta B} = \{3, 4, 5, 7\}$$

$$\overline{A \Delta B} \cap C = \{5, 7\}$$

$$A = \{1, 3, 5\}$$

$$B = \{2, 3, 5, 6\}$$

$$C = \{5, 6, 7\}$$

$$(A \cap B) \times \mathcal{P}(B - C)$$

$$A \cap B = \{3, 5\}$$

$$B - C = \{2, 3\}$$

$$\mathcal{P}(B - C) = \{\emptyset, \{2\}, \{3\}, \{2, 3\}\}$$

$$(A \cap B) \times \mathcal{P}(B - C)$$

$$= \{ \{3, \emptyset\}, \{3, \{2\}\}, \{3, \{3\}\},$$

$$\{3, \{2, 3\}\}, \{5, \emptyset\}, \{5, \{2\},$$

$$\{5, \{3\}\}, \{5, \{2, 3\}\} \}$$

1) QUTS ✓

2) H-DS ✓

3) P-DS (R1S)

4) TE-DS H

1. T (T1TQ)

TTUQ

TT

Q

5) TE-DS SH 2, 8

6) SV (R1S) LC 3, 5

7) S Absorption 6

8) Q SD 2, 7

9) QUTT, Adición 8

10) TTUQ, con 9

11) T (T1TQ)

$\neg P \vee \neg R$

1) $P \vee Q \checkmark$

2) $\neg R \vee \neg P \checkmark$

3) $S \rightarrow R$

4) $T \vee S$

5) $R \rightarrow \neg Q \checkmark$

T

T1?

6) $\neg P \rightarrow \neg Q$ ID 1

7) $\neg P \vee \neg R$, Conm, 2

8) $P \rightarrow \neg R$, ID 7

9) $Q \vee \neg R$, LC, 6, 8

10) $\neg R \vee \neg Q$, ID 5

11) $\neg Q \vee \neg R$, Con, 10

12) $Q \rightarrow \neg R$, ID 11

13) $\neg Q \rightarrow \neg R$, ID 9

14) $\neg R \vee \neg R$, LC 13, 12

15) $\neg R$, Idem 14

16) $\neg S$, mt 15, 3

17) T

$$1) Q \rightarrow S \checkmark$$

$$5) \neg P \rightarrow S, SH \text{ 1, 2}$$

$$2) \neg P \rightarrow Q \checkmark$$

$$6) \underline{S \vee (R \wedge S)}, LC \text{ 3, 5}$$

$$3) P \rightarrow D(R \wedge S)$$

$$7) S, ABS \text{ 6}$$

$$4) \neg A \vee \neg S$$

$$8) \neg A, SD \text{ 7, 4}$$

$$\therefore, T \rightarrow \neg \neg A$$

$$9) \neg T \vee \neg A, Addition \text{ 8}$$

$$\neg T \vee \neg A$$

$$10) T \rightarrow \neg \neg A, ID$$

$$\neg T$$

$$\neg A$$